

Multi-Head Spectral Anchor Pooling for Protein Language Model Representations in Antimicrobial Peptide Activity-Cliff Regression

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March 2026

Abstract

We introduce **multi-head spectral anchor pooling**, a sequence-to-vector module for fine-tuning protein language models (PLMs) on regression over antimicrobial peptide (AMP) *activity cliffs*. Pooling is difficult because cliff labels depend on subtle sequence edits; standard mean, max, or learned attention pooling may under-use structure in the token trajectory. Our module optionally maps hidden states along the sequence dimension with an **FFT-based linear transform** (real and imaginary parts of `rfft`), splits representations into **multiple heads**, and assigns each position a weight from **learnable anchor prototypes** with **per-head temperature**. Crucially, weights are inferred from (optional) spectral features, but **aggregation uses the original time-domain token representations**, decoupling “where to attend” from “what to pool.” An optional **sigmoid gate** and output projection follow head concatenation. Under a fixed **ESM-2-t12** backbone and a fixed activity-cliff split (`diff=5`) on *E. coli* and *S. aureus* datasets, we compare against mean, max, attention pooling, and single-stream **spectral_anchor**. Training uses `model.regression.apply=None` (no DC filtering in this study). On *E. coli* test Spearman correlation, the best multi-head configuration reaches **0.558** versus **0.555** for attention pooling and **0.504** for mean pooling. On *S. aureus*, the best multi-head configuration reaches **0.453** versus **0.406** for attention pooling; **test metrics are missing** for the exported mean/max baselines in our metrics sheet and must be re-run before claiming a full baseline table.

1 Introduction

Predicting whether a small sequence change induces a large potency jump—an *activity cliff*—is central to peptide lead optimization. PLMs such as ESM-2 provide contextual embeddings per residue, but downstream models require a **single vector** per sequence. Common poolings (mean, max, attention) treat the trajectory uniformly or learn content-based weights entirely in the time domain, which may not align with periodic or multi-scale structure along the sequence.

We propose **multi-head spectral anchor pooling** with three design goals: (i) an **optional harmonic decomposition** along sequence length as an inductive bias complementary to the PLM’s implicit modeling; (ii) **prototype-style** assignment to learnable anchors; (iii) **multi-head subspaces** with independent temperatures and anchor sets.

Contributions.

- A pooling layer that combines **FFT-feature-driven weights** with **time-domain aggregation** over the original hidden states.

- A **multi-head** extension with orthogonal anchor initialization, per-head softmax temperatures, optional gating, and Hydra wiring `num_anchor_per_head = num_anchor // num_heads` when `num_anchor >= num_heads`, else 2 (see `factory/pooling/registry.py`).
- Empirical results on AMP activity-cliff regression (ESM-2-t12, two organisms) against mean / max / attention / `spectral_anchor`, plus ablations over `num_heads`, `use_fft`, and `gated`, from `outputs/ablation_metrics_exports/ablation_metrics_summary.csv`.

We **do not** report DC-component processing (`filter_dc`, `scale_dc`, etc.); the ablation driver sets `model.regression.apply=none` (`evaluation_scripts/run_spectral_anchor_v2_ablation.sh`).

2 Method

2.1 Backbone and task

Given a sequence, a PLM yields hidden states $X \in \mathbb{R}^{B \times T \times d}$ and a mask over valid positions. A regression head predicts a scalar target. Only the pooling operator changes across runs.

2.2 Multi-head spectral anchor pooling

Optional spectral path. If `use_fft` is true, compute $Z = \text{rfft}(X)$ along the sequence axis. Concatenate real and imaginary parts along the feature axis, truncate or pad to width $2d$, and apply a learned linear map $W_s : \mathbb{R}^{2d} \rightarrow \mathbb{R}^d$ to obtain H . If `use_fft` is false, $H = X$.

Multi-head split. Reshape H into H_{heads} subspaces of dimension $d_h = d/H_{\text{heads}}$ per head (implementation uses `view` after FFT; sequence length may be $T' = \lfloor T/2 \rfloor + 1$).

Anchor assignment. Per head h , learn anchors $A_h \in \mathbb{R}^{K \times d_h}$ (orthogonal init). Euclidean distances yield softmax weights over anchors with learnable temperature τ_h ; a **mean** over anchor indices yields one scalar weight per time step for that head.

Align length and aggregate. Interpolate weights to length T if needed, apply the mask, renormalize. For each head, **pool the original** X slice: $p_h = \sum_t w_{t,h} \odot x_{t,h}$. Concatenate heads, optionally apply $\sigma(W_{gp}) \odot p$, then a linear output projection.

2.3 Configuration

The ablation script sweeps `num_heads` $\in \{2, 4, 8\}$, `use_fft` $\in \{\text{true}, \text{false}\}$, `gated` $\in \{\text{true}, \text{false}\}$, with default `num_anchor=8` unless overridden.

2.4 Pseudocode

```
Input: X (B,T,d), mask M
if use_fft:
    Z = rfft(X, dim=sequence)
    S = concat(Re(Z), Im(Z)); F = Linear_{2d->d}(S)
else: F = X
split F into H heads -> per-head anchor softmax -> weights W_h
interpolate W_h to length T; normalize(W_h * M)
```

```

for each head h: p_h = sum_t W_h[t] * X[t, head h]
p = concat(p_h); optional gate; return Linear_out(p)

```

3 Experiments

3.1 Setup

Model: ESM-2-t12. Data: activity-cliff splits for *E. coli* and *S. aureus*, `diff=5`, fixed CSVs. Metric: test Spearman from the exported CSV. Training aligns with `run_spectral_anchor_v2_ablation.sh` (`apply=none`).

3.2 Main comparison

Table 1: Test Spearman (ESM-2-t12, `spectral_anchor_v2_ablation` rows, `diff=5`).

Pooling	<i>E. coli</i>	<i>S. aureus</i>
mean	0.5038	— (missing in export)
max	0.5525	— (missing in export)
attn	0.5547	0.4060
multi_head (best)	0.5584	0.4531

Best multi-head settings: *E. coli*: $H=8$, FFT on, gated on. *S. aureus*: $H=2$, FFT on, gated off.

3.3 Ablations

Table 2: *E. coli*: multi-head spectral, test Spearman.

H	FFT	Gated	Test Spearman
2	off	off	0.5405
2	off	on	0.5262
2	on	off	0.5182
2	on	on	0.5052
4	off	off	0.5367
4	off	on	0.5445
4	on	off	0.5493
4	on	on	0.5499
8	off	off	0.5358
8	off	on	0.5425
8	on	off	0.5579
8	on	on	0.5584

4 Related Work

We cite only verified bibliography later; placeholders: PLMs [?]; Fourier layers [?, ?, ?]; set pooling [?, ?]; multi-head attention [?]; gating [?, ?]; non-local weighting [?].

Table 3: *S. aureus*: multi-head spectral, test Spearman.

<i>H</i>	FFT	Gated	Test Spearman
2	off	off	0.4130
2	off	on	0.4256
2	on	off	0.4531
2	on	on	0.4559
4	off	off	0.4285
4	off	on	0.4443
4	on	off	0.4372
4	on	on	0.4293
8	off	off	0.4197
8	off	on	0.4206
8	on	off	0.3971
8	on	on	0.4247

5 Limitations

Incomplete *S. aureus* mean/max test rows in CSV; single-seed runs; no DC pipelines in this draft; frequency–motif interpretability and formal theory left to future work.

References

- [1] *[PLACEHOLDER]* Protein language model (e.g., ESM-2) — verify authors, title, venue, DOI before submission.
- [2] *[PLACEHOLDER]* FNet (Lee-Thorp et al.) — verify.
- [3] *[PLACEHOLDER]* AFNO — verify.
- [4] *[PLACEHOLDER]* FEDformer — verify.
- [5] *[PLACEHOLDER]* Set Transformer — verify.
- [6] *[PLACEHOLDER]* Deep Sets — verify.
- [7] *[PLACEHOLDER]* Vaswani et al., Transformer — verify.
- [8] *[PLACEHOLDER]* GLU (Dauphin et al.) — verify.
- [9] *[PLACEHOLDER]* SENet — verify.
- [10] *[PLACEHOLDER]* Non-local Neural Networks — verify.